

Maya Chen

Data analyst exploring backend development

SYNTHETIC DEMO MATERIAL | All people, projects and results are fictional.

Profile

Fictional analyst transitioning toward software development. Has project evidence in SQL, Python data validation and stakeholder communication, with limited API and PostgreSQL experience.

Skills

Python, pandas, SQL joins and aggregation, SQLite, Git basics, data validation and documentation.

Inventory reconciliation exercise

Compared two synthetic inventory exports and identified duplicated product identifiers, missing rows and inconsistent date formats.

Created a Python validation script that separates invalid records and produces a summary of corrections. Added eight tests for missing fields, duplicate IDs and malformed dates.

Used SQLite for local analysis. Did not implement a multi-user PostgreSQL service, HTTP authentication or concurrent updates.

Reporting project

Built weekly SQL summaries of orders and stock levels using generated data. Checked totals against a hand-calculated sample before presenting the results.

Discussed inconsistent definitions of an active order with fictional operations staff and documented one agreed definition. No performance benchmark is claimed.

Development goals

Currently learning HTTP request handling, REST conventions and automated API tests. No deployed backend service or commercial engineering experience is claimed.

Learning background

Fictional coursework in introductory programming, statistics and database queries. The projects provide limited evidence for API design and concurrent transaction handling.